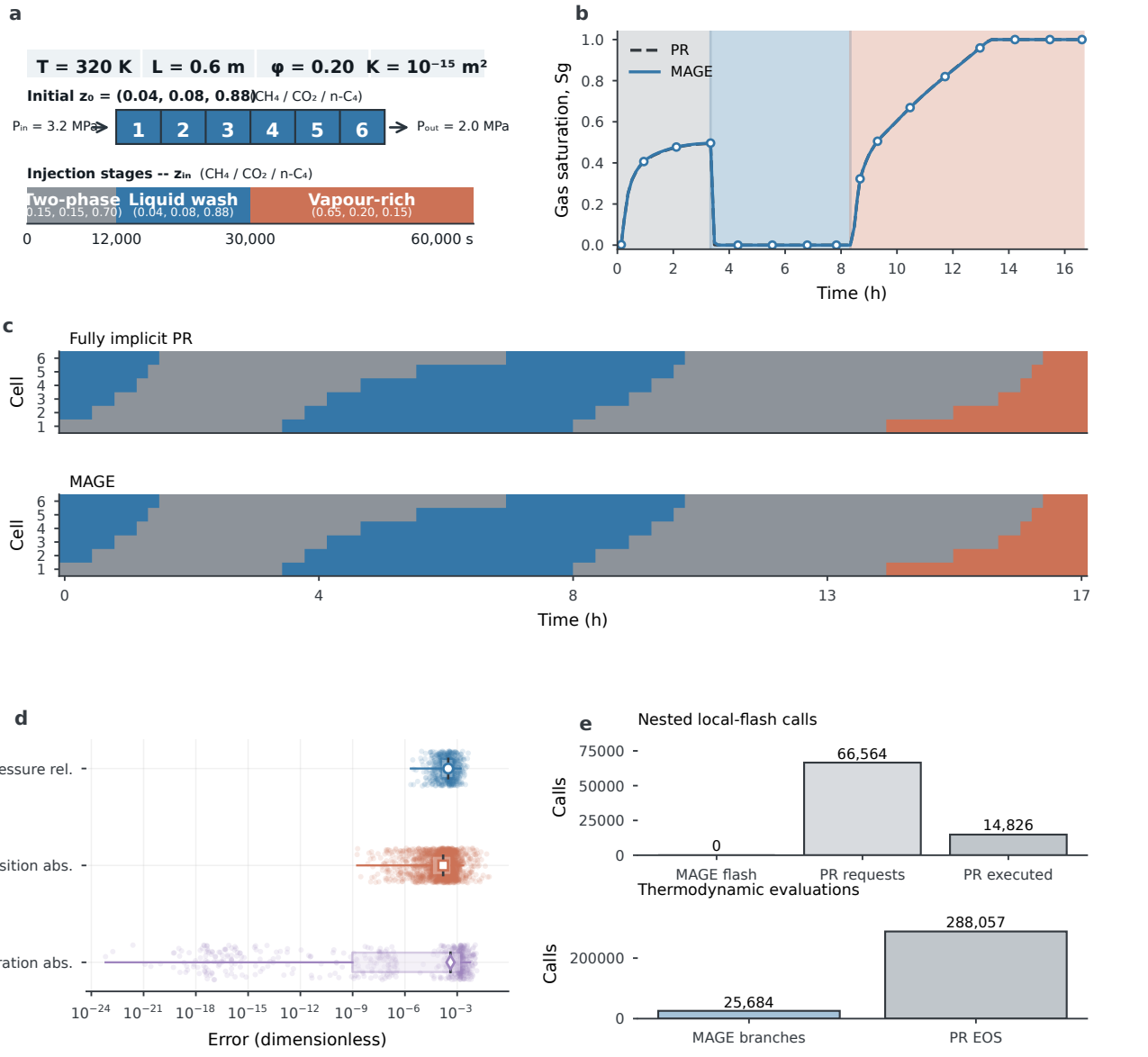

a
T = 320 K
L = 0.6 m
φ = 0.20
K = 10⁻¹⁵ m²
Initial z₀ = (0.04, 0.08, 0.88) (CH₄ / CO₂ / n-C₄)
Pin = 3.2 MPa
1
2
3
4
5
6
Pout = 2.0 MPa
Injection stages -- zin (CH₄ / CO₂ / n-C₄)
Two-phase
(0.15, 0.15, 0.70)
Liquid wash
(0.04, 0.08, 0.88)
Vapour-rich
(0.65, 0.20, 0.15)
0
12,000
30,000
60,000 s
b
PR
MAGE
Gas saturation, Sg
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
12
14
16
Time (h)
c
Fully implicit PR
Cell
6
5
4
3
2
1
MAGE
Cell
0
4
8
13
17
Time (h)
d
Pressure rel.
Composition abs.
Saturation abs.
10⁻²⁴
10⁻²¹
10⁻¹⁸
10⁻¹⁵
10⁻¹²
10⁻⁹
10⁻⁶
10⁻³
Error (dimensionless)
e
Nested local-flash calls
Calls
75000
50000
25000
0
66,564
14,826
0
MAGE flash
PR requests
PR executed
Thermodynamic evaluations
Calls
200000
0
288,057
25,684
MAGE branches
PR EOS